<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks Available</th><th rowspan="2">Maths</th><th rowspan="2">Prac</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th></tr>
<tr><td>5 FT</td><td>(a)</td><td></td><td></td><td>Indicative content:
Energy will transfer from the warm house to the cold surroundings.
The uPVC is a better insulator than the aluminium. This will reduce the heat loss via conduction. Also the white surface is a poor emitter of infra-red, so reduces losses by radiation.
The reflective lining is a good reflector and poor emitter of heat radiation / infra-red and will reduce heat loss via radiation.
The vacuum between the glass panes will prevent both conduction and convection losses as there are no particles and the reduced temperature on the outside of the glass reduces convection losses outside the house.

5 – 6 marks
Detailed description of how heat transfer is reduced by conduction, convection and radiation.
There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.

3 – 4 marks
Detailed description of how heat transfer is reduced by 2 out of conduction, convection and radiation. Alternatively, a brief correct description of all 3 methods.
There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.

1-2 marks
Detailed description of how heat transfer is reduced by 1 out of conduction, convection and radiation. Alternatively, a brief correct description of up to 2 of the methods.
There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</td><td>3</td><td>3</td><td></td><td>6</td><td></td><td></td></tr>
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Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				Note AO1 marks are for descriptions of heat transfer processes whereas the AO2 marks are for application of knowledge of how the features shown reduce transfer by those methods. **0 marks** *No attempt made or no response worthy of credit.*						
	(b)	(i)		[Payback = $\frac{4\,000}{80}$] = 50 [years]		1		1	1	
		(ii)		Cost of electricity / gas changes / cost of a unit changes (1) the number of units used changes / amount of electricity or gas used changes (1) **Award 1 mark only** for this part of the question for answer of the heating bill may change **Award 1 mark only** for annual savings will change **Don't accept** costs will change / more or less heating will be used		2		2		
				Question 5 total	**3**	**6**	**0**	**9**	**1**	**0**

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